

Probability

Random Variables, Probability Distributions

Stat 801a

Population versus Sample

Population – The set of all objects under consideration. Objects in a population are called *elements*.

Examples

- Boxes of a particular brand of corn flakes.
- Newly made light bulbs in a warehouse at a given time.

Another Definition of Population

- All possible outcomes of an *experiment* can also be values in a population.
- Flipping a coin twice. Possible outcomes are: HH, HT, TH, TT. Could make the outcomes numeric using 3, 2, 1, 0 to be the elements corresponding to the above events.
- If the experiment is repeated many times, the resulting values constitute a population: 2, 0, 3, 1, 0, 2, ...

Random Sampling

- Sample – A subset of the elements from a population.
- Simple Random Sample of size n from a finite population is a subset selected in such a way that every subset of n elements is equally likely to be the one selected. (Abbreviation: SRS)
- A random sample from a small finite population can be obtained using the Random Numbers - using code or random number table.

- sample(1:100,3) or sample(3)
- Computer programs are used to draw samples from large populations.

Probability, Random Variables, and Probability Distributions

- Assume a numeric finite population of N elements, and suppose n_1 of the elements are the number k .
- Suppose a SRS of size 1 is taken from this population.
- The probability that the number k is selected is n_1/N .
- That is, we define this probability to be the relative frequency of occurrence of k in the population.

Probability: Example

- Example: $N = 10$
- Population: 1, 1, 1, 3, 7, 7, 9, 9, 9, 9
- Denote by $P(k)$, the probability of the number k being selected
- $P(1) = 3/10$, $P(3) = 1/10$, $P(7) = 2/10$,
 $P(9) = 4/10$, $P(2) = 0/10$, $P(1 \text{ or } 3) = 4/10$,
 $P(\text{not } 7) = 1 - 2/10$.

Random Variables

A random variable is a function defined on a numeric population. We will often say that this function value is the value assumed by the random variable.

- Example: $N = 10$
- Population: 1, 1, 1, 3, 7, 7, 9, 9, 9, 9
- Let X be a random variable defined on this population.
- X can assume any of the values 1, 3, 7, or 9.

Random Variables (continued)

- A Random variables are different from ordinary variables because they assume values from the population with associated probabilities.
- For e.g., we will say
$$P(X = 1) = 3/10, \quad P(X = 7) = 2/10,$$
$$P(X = 2) = 0/10, \quad P(X = -6) = 0/10,$$
$$P(1 \leq X \leq 3) = 4/10, \quad P(1 \leq X \leq 9) = 1. \text{ etc.}$$
- A probability distribution specifies the probabilities associated with all possible values the random variable X can assume.

The M & M Example

Example:

If X represents the number of RED *M&Ms* in a vending machine size package, then perhaps the probability distribution of X is something like the following:

x	5	6	7	8	9	10	11
$P(X = x)$	$\frac{2}{28}$	$\frac{2}{28}$	$\frac{5}{28}$	$\frac{7}{28}$	$\frac{5}{28}$	$\frac{4}{28}$	$\frac{3}{28}$

The M & M Example (continued)

We read this table by finding a value of X , say 8, then reading the probability below it:

- The probability of observing 8 M&Ms in a package is $\frac{7}{28}$, meaning, if we open many many packages of M&Ms, about 7 out of 28 will have exactly 8 M&Ms within.

This is an example of a discrete random variable.

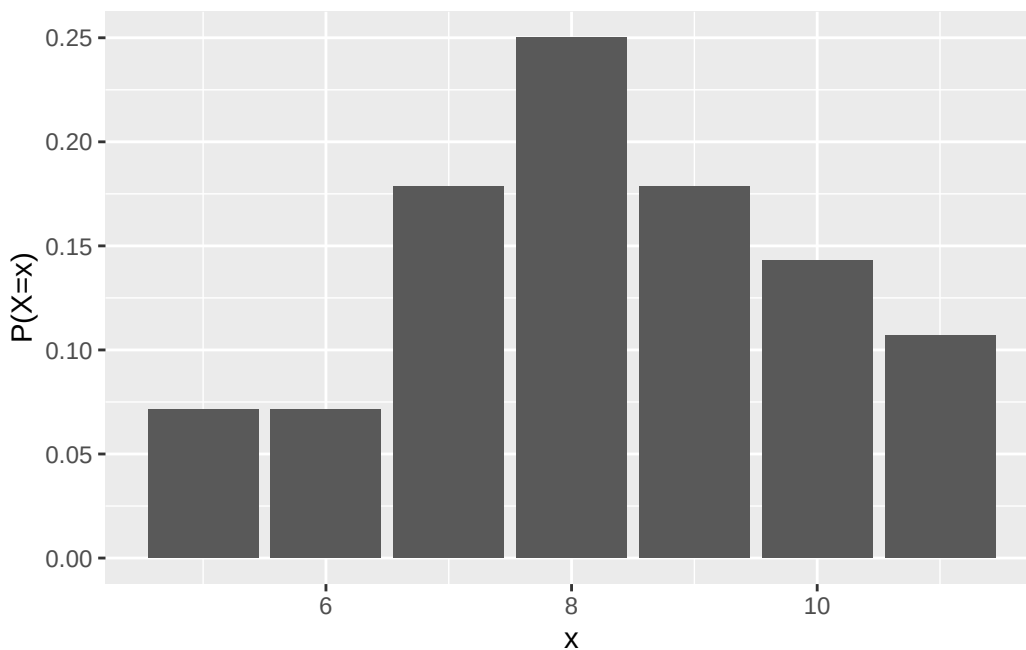
Discrete Random Variables

- When the elements of a population consists of a countable discrete set of values the random variable defined on that population is called discrete random variable.
- For discrete random variables the probability distribution can be displayed using a histogram or a table of values and associated probabilities as shown above.

Properties of Probability Distributions of Discrete Random Variable X

- $0 \leq P(X = x) \leq 1$ for all x
- $\sum_x P(X = x) = 1$

The M & M Probability Distribution



Continuous Random Variables

- When the population consists of an uncountable number of values in a line interval use integration to calculate proportions of the population that exist in any given subinterval of a continuum.
- For a random variable defined on such a population we will deal with the probability that the random variable assumes a value in a given interval, not that it assumes a specific numeric value.

Continuous Random Variables (contd.)

- When the population consists of a set of values in a continuous range the random variable defined on that population is called continuous random variable.

- The probability distribution of a continuous random variable is a smooth curve.
- The area under the curve between specified values of the random variable correspond to a probability.
- The probability that a continuous random variable assumes a specific numeric value is zero.

Continuous Random Variables (contd.)

- Example: A Continuous Random Variable X defined on
- a Population: all numbers in the interval $(0,2)$ occurring with various frequencies (that we won't specify here).
- By Definition

$$P(1/2 \leq X \leq 1/2) \equiv P(X = 1/2) = 0$$

$$P(0 \leq X \leq 2) = 1, \quad P(X < 0) = P(X \leq 0) = 0$$
- We could possibly calculate the following probability

$$P(1 \leq X \leq 2) = ?$$

Binomial Random Variable

- For some interesting populations that occur naturally, theoretical probability distributions and associated random variables, have been defined.
- One such population features elements that are only one of two kinds (Yes or No; Good or Defective; Below or Above;, etc.).
- Let us use Success(S) and Failure(F) to designate population elements, in general
- Consider a typical population of this kind consisting of m S's and $N - m$ F's, where N is the total number of elements in the population.

Binomial Experiment

- A binomial experiment consists of taking n successive SRS, each of size 1, replacing elements between draws, and counting the number of S's obtained.
- The binomial experiment is conducted by sampling from the population of S's and F's.
- The proportion of S's in the population is $\pi = m/N$ so the proportion of F's is $1 - \pi \equiv (N - m)/N$.

- Let X_i be the random variable that assumes the value of the outcome of the i -th draw for each $i = 1, 2, \dots, n$.

Binomial Distribution

- Let the random variable X_i assume the value 1 if the outcome of the i -th draw is S or the value 0 if the outcome is F.
- Thus we imagine n random variables, $X_1, X_2, X_3, \dots, X_n$ each assuming a value S with probability π , or F with probability $1 - \pi$.
- That is, the probability distribution of *each* of the n random variables $X_1, X_2, X_3, \dots, X_n$ is given by:

$$P(X = x) = \begin{matrix} \overline{x} & 0 & 1 \\ \overline{1 - \pi} & \pi & \pi \end{matrix}$$

Binomial Distribution (contd.)

- Now define a new random variable Y so that

$$Y = X_1 + X_2 + X_3 + \dots + X_n$$

- Y represents the number of Success(S)'s in the set of values assumed by $X_1, X_2, \dots, X_n$.
- This is because each $X_i = 1$ when an S is drawn, and $X_i = 0$ otherwise, we can see that Y represents the total number of S's in the experiment.
- Y is said to be a binomial random variable.

Binomial Distribution (contd.)

- That is, a binomial random variable assumes the value of the total number of successful outcomes in a binomial experiment.